

Muhammad Alfath Abibi

Software Engineer

alfath.abibi@gmail.com (+62)82343666626 Bandung, Indonesia

<https://www.linkedin.com/in/muhammad-alfath-a-b691901b7/>

🔗 Summary

Full Stack Engineer with experience delivering enterprise and government systems across Indonesia and Singapore. Strong background in backend development (.NET, Java Spring, NestJS), frontend frameworks (Vue.js, Razor Pages, Svelte), cloud platforms (AWS, GCP), and quality engineering (unit testing, SonarQube, regression testing). Proven ability to work on large-scale, regulated systems with high reliability and test coverage requirements.

🏢 Experience

Software Development Engineer – The Software Practice

2022-11-01 - Present

Enterprise Transportation Operations System

(Government Transportation Sector – Singapore)

- Developed and maintained an enterprise-grade operations system for a national transportation organization
- Implemented features for: Shift scheduling and exchange, Leave management, Digital shift handover with signature, Operational balance tracking across multiple locations.
- Built frontend using **Vue.js**
- Developed backend services using **.NET**

Tech: Vue.js, .NET

National Health Product Registration Platform

(Public Health Regulatory System – Singapore)

- Worked as **Frontend Engineer** on a large-scale regulatory web platform
- Implemented UI strictly aligned with **Figma design system**
- Integrated frontend with backend workflows for product registration and application processing
- Resolved **SonarQube issues** and maintained **≥80% unit test coverage**
- Developed **regression test suites using Playwright** based on business scenarios
- Implemented secure file upload architecture using: **AWS S3** for object storage, **AWS Lambda** as upload endpoint

Tech: Razor Pages, Svelte, Playwright, AWS S3, AWS Lambda

Government Land Administration System

(Public Sector Land Management Platform – Singapore)

- Developed frontend modules using **Razor Pages + Alpine.js**
- Addressed SonarQube issues across frontend and backend codebases
- Built and maintained **unit tests with ≥80% coverage**
- Supported a multi-portal system serving: Government officers, Other public agencies, Corporate and individual users
- System used for land-related application submission and case management

Tech: Razor Pages, Alpine.js, .NET, AWS S3

Backend Developer – Hijrah Coach 2022-06-01 - 2023-08-01

- Built a platform connecting **business coaches and entrepreneurs**
- Developed AI-powered business consultation features using internal datasets
- Backend services implemented using **NestJS**
- Implemented **real-time video and voice communication** using **Socket.IO**
- Used **Google Cloud Platform** for file storage (images, PDFs)
- Researched and evaluated **cost-effective AI tools** aligned with business constraints

Tech: NestJS, Socket.IO, Google Cloud Platform, AI Tools

Backend Developer – PT Icon Plus Indonesia 2021-09-01 - 2022-05-01

- Developed backend services for **Yanbung**, used by **PLN officers nationwide**
- Handled workflows for: New electricity meter requests, Meter upgrades, Temporary meter release for public events
- Built backend using **Java Spring Boot**
- Implemented **Kafka** for asynchronous processing and messaging

Tech: Java Spring Boot, Kafka

🎓 Education

Diploma, Telkom University (3.93/4.00)	2021-09-14
Bachelor, Telkom University (3.80/4.00)	2023-08-20

🔧 Project

Smart Hydroponic IoT System (Academic)
Full Stack Engineer

- Developed an IoT-based hydroponic automation system to control **water pH and temperature**
- Enabled real-time monitoring and control through an **Android application**
- Implemented **Firebase Realtime Database** for live data synchronization
- Used **Arduino** as sensor controller and actuator trigger for pumps and nutrient control
- Recognized as **Top 10 Winner nationally** in Telkom Indonesia Innovillage competition

Tech: Arduino, Android, Firebase Realtime DB, IoT Sensors

Labaco (Startup – Side Project)

Full Stack Engineer

- Developed a ride-hailing platform for underserved regional areas
- Built backend and web frontend using **CodeIgniter**
- Developed mobile applications using **Flutter**
- Integrated third-party services: Google Maps & Places API, Firebase (chat & push notifications), Xendit payment gateway (multi-payment support)
- Optimized map costs by: Using **Mapbox for routing (polyline)**, Rendering routes on Google Maps
- Application was published on **Google Play Store and App Store**

Tech: CodeIgniter, Flutter, Firebase, Google Maps, Mapbox, Xendit

☆ Technical Skills

Backend:

.NET, Java Spring Boot, NestJS, Kafka

Frontend:

Vue.js, React.js, Razor Pages, Svelte, Alpine.js

Mobile:

Flutter

Cloud & Infrastructure:

AWS (S3, Lambda), Google Cloud Platform

Testing & Quality:

Unit Testing, SonarQube, Playwright (Regression Testing)

Others:

Socket.IO, Firebase, Payment Gateway Integration, IoT Systems